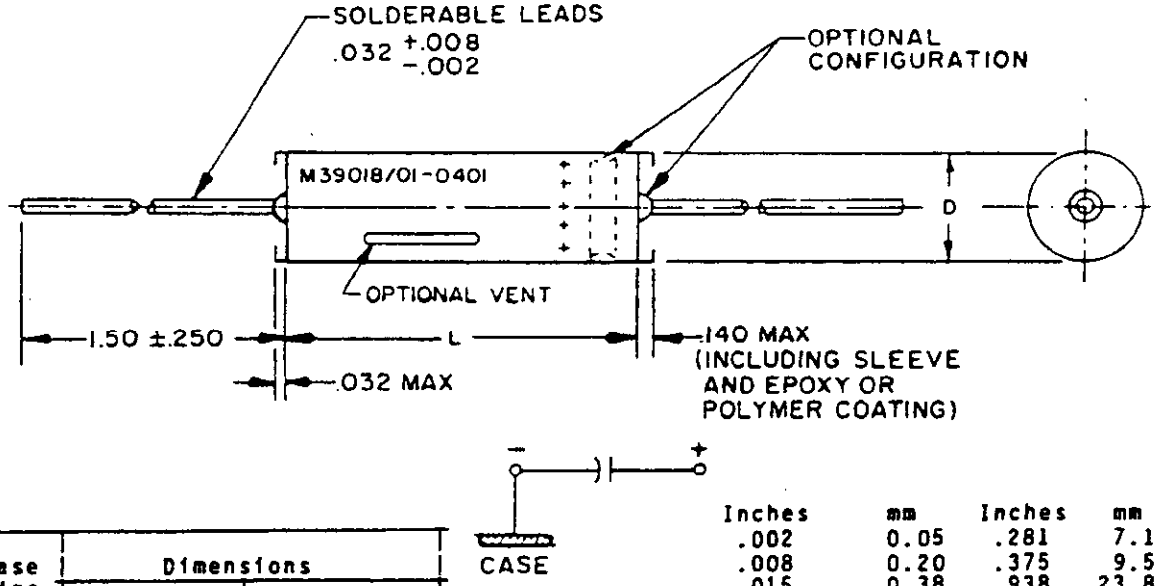
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE) (POLARIZED), NON-ESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY, STYLES CU12 (UNINSULATED), CU13 (INSULATED), AND CUR13 (INSULATED)

This amendment forms a part of MIL-C-39018/1D, dated 23 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Figure 1, delete and substitute:



Case size	Dimensions	
	L ± .031	D ± .015 See note 4
A0	.938	.281
A1	.938	.375
A2	1.125	.375
A3	1.375	.375
A4	1.625	.375
A5	2.188	.375
A6	2.688	.375

Inches	mm	Inches	mm
.002	0.05	.281	7.14
.008	0.20	.375	9.52
.015	0.38	.938	23.83
.031	0.79	1.125	28.57
.032	0.81	1.375	34.92
.140	3.56	1.50	38.1
.250	6.35	1.625	41.27
		2.188	55.58
		2.688	68.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These capacitors are not intended to be mounted by their leads.
4. The "D" dimension for styles CU13 and CUR13 shall be not more than .008 (0.20 mm) larger than the specified bare case diameter.

FIGURE 1. Styles CU12, CU13, and CUR13 capacitors.

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PAGES 2 AND 3

Table I, dash number, style CU13, capacitance tolerance headings: Delete "-10 -30", "-10 -50", and "-10 -75", and substitute "-10 +30", "-10 +50", and "-10 +75", respectively.

PAGE 3

Table I, footnote 2: Add "For CUR13 with random vibration, the letter R shall be used in place of the dash (i.e., M39018/01R1046M).".

PAGE 5

Following vibration, high frequency: Add "Random (style CUR13 only) (when applicable) - in accordance with MIL-C-39018."

PAGE 6

Marking, part number: Add "For style CUR13 parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/01R1046M).".

PAGE 8

Add the following application note:

"APPLICATION NOTE:

Ripple current: The maximum permissible ripple current for these capacitors can be determined from table III. The appropriate multiplier from this table should be used where operation at other frequencies or temperatures is desired. Operation of these capacitors outside the limits of 60 to 100,000 Hz is not recommended.

TABLE III. Ripple current multipliers.

With temperature	+25°C	+45°C	+65°C	+85°C	+125°C
	1.4	1.4	1.2	1.0	0.3
With frequency		60 Hz	120 Hz	400 Hz	1 K through 100 K Hz
	0-60 volts	.85	1.0	1.1	1.15
	61-200	.83	1.0	1.15	1.2
	201-350	.80	1.0	1.3	1.4

"

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5910-1688-1)

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